



Evaluating the impact of filter-based feature selection in intrusion detection systems

Houssam Zouhri¹ · Ali Idri^{1,2} · Ahmed Ratnani¹

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Abstract

High dimensionality can lead to overfitting and affect the modeling power of classification algorithms, resulting an increase in false positive rate (FPR) and false negative rate (FNR). Therefore, feature selection is a critical issue to deal with by means of efficient techniques when developing Intrusion Detection Systems. This study seeks to assess and compare the impacts of five univariate filters (Relieff, Pearson Correlation, Mutual Information, ANOVA, and Chi2) with different selection thresholds, and three multivariate filters (Correlation-based feature subset selection, Double Input Symmetric Relevance and Consistency-based subset selection) on the performances of four classifiers (Multilayer Perceptron, Support Vector Machines, XGBoost and Random Forest) over CIC-IDS2017, CSE-CIC-IDS2018 and CIC-ToN-IoT intrusion detection datasets. We evaluate 228 variants of classifiers to determine the features that positively impact the classification efficiency of all the cyber-attack scenarios used. The obtained results show that using XGBoost and Random Forest trained with multivariate methods, such as CON and DISR, can effectively reduce the number of features without affecting the classification performance and detection rate, compared to other filtering methods.

Keywords High dimensionality · Intrusion detection systems · Feature selection · Univariate filters · Multivariate filters · Cyber-attack

1 Introduction

Intrusion detection consists of monitoring and analyzing activities to detect and alert on unauthorized events that violate security rules or standard security practices on computer systems, networks, or other system's information [1]. An intrusion can occur in several ways, including stealing someone's password, gaining illegal access to a system or network, and impersonating a valid user in an attempt to access illegal data. The Intrusion Detection System (IDS) is a crucial component for network security since it detects mali-

cious data entering the network, raises an alarm, and takes appropriate action [2]. IDSs are classified into two types: host-based IDS (HIDS) and network-based IDS (NIDS). HIDSs monitor the activity of a specific host or device and are typically installed on servers, workstations or other individual devices. NIDSs keep an eye on network traffic and are placed at key points on the network.

In addition, IDSs can operate in two modes: rule-based systems and signature-based systems. In rule-based system, also known as anomaly-based IDS, data are collected about normal network activities and used to build a model of what a normal activity looks; this allows to detect new attacks in the system without any prior knowledge. The rule-based system checks data with the already stored behavior states in the IDS database, and if it senses any abnormality, which means a deviation from the specified rules, an alarm is generated to indicate the presence of an intrusion in the network. The main benefit of anomaly-based system is that they can continuously learn and adapt to changes in normal behavior, making them effective against new and unknown threats [3, 4]. Despite their advantages, they have high false alarm rate and cannot provide explanation for anomalies [5]. For

✉ Ali Idri
ali.idri@um6p.ma

Houssam Zouhri
houssam.zouhri@um6p.ma

Ahmed Ratnani
Ahmed.Ratnani@um6p.ma

¹ Mohammed VI Polytechnic University (UM6P), Benguerir, Morocco

² Software Project Management Research Team, Mohammed V University, Rabat, Morocco

signature-based IDS, also called misuse detection-based systems, it consists of comparing activity or system logs against a database of known threat signatures. In this mode, when the IDS system detects a match between incoming data and a known threat signature, it triggers an alert and takes appropriate action, such as blocking traffic or alerting security personnel. Some advantages of using this system include: the efficiency and speed of the known threats detection, as they are specifically designed to recognize specific patterns of malicious activity, and they do not need to analyze or learn normal behavior before identifying an attack [6]. However, the major drawback of these signature-based IDSs is their inability to detect an attack if it is not found in the IDS database.

According to a survey on the utilization of existing feature selection (FS) algorithms in IDSs [7], several researchers have proposed identifying important intrusion features as a preprocessing task to enhance the learning model's performance and make it more computationally efficient. The aim of FS is to choose a subset of the most relevant features or variables that can minimize computation and complexity while improving model performance and interpretability. FS can help improve the interpretability difficult-to-understand black-box models, especially for domain experts who are unfamiliar with the technical details of the model. The survey [7] reviewed a set of 170 primary papers, published during the period 2008–2020 in the field of IDS, from different perspectives including FS techniques, feature engineering types and datasets used to evaluate the IDSs performances. This survey highlighted the use of various FS techniques over different IDS datasets and outlined that the experimental design of the most reviewed studies was not complete since it did not include any comparative study of different univariate and multivariate filters. The comparative study [8] evaluated the impact of three different FS algorithms (Recursive Feature Elimination, Information Gain, and Chi-Square) on the performance of various machine learning classifiers including Naïve Bayes (NB), Decision Tree Classifier (DT), Support Vector Machine (SVM), Random Forest (RF), K-Nearest Neighbors (KNN), Artificial Neural Networks (ANN) and Logistic Regression (LR) using NSL-KDD dataset. The findings showed that IDS model performance was enhanced when incorporating FS methods into the model, and the combination of the Support Vector Machine classifier and Recursive Feature Elimination yielded comparatively better results for the classification of DoS, R2L, U2R and Probe attacks, with accuracies of 99.6%, 98.9%, 98.2% and 99.1%, respectively. Furthermore, while extensive surveys and studies have contributed significantly to the field of IDSs, several notable limitations persist. These surveys [7, 8] have revealed some critical gaps in the existing body of research, motivating the need for our study: (1) There is no primary study that compared the impacts of filters on the classification

performance in the cybersecurity domain. (2) Most studies have been conducted on old datasets, which may limit the effectiveness and applicability of the study results. And (3) the majority of previous studies prioritized improving classification accuracy and ignored the interpretability of the classifiers used, which were primarily black box models. Our study is driven by the motivation to address these limitations by conducting an empirical evaluation of FS techniques and ML algorithms, their interactions, and their generalizability across diverse network environments. Through our research, we aim to bridge these gaps in the literature, providing actionable insights to enhance the effectiveness and interpretability of IDSs, by identifying the pivotal features influencing the classification of intrusions using different attack scenarios, ultimately contributing to the advancement of network security.

Therefore, the purpose of this research is to address these limitations by evaluating and comparing the impacts of five univariate filters: Pearson correlation [9], Relief [10, 11], Mutual Information [12], Anova [13] and Chi2 [14] and three multivariate filters: Consistency-based subset selection [15], Correlation-based feature subset selection [16], and Double Input Symmetric Relevance [17] on the performance of four classifiers: Multilayer Perceptron (MLP), SVM, XGBoost (XGB) and RF) over the three recent intrusion detection datasets CIC-IDS2017, CSE-CIC-IDS2018 and CIC-ToN-IoT. The rationale behind choosing the aforementioned filter techniques is that they are the most widely used filters for selecting the best feature subset in several areas, such as heart disease classification [18], software effort estimation [19], educational data mining [20], text classification [21] and IDSs [7, 8],[22–25]. In addition, we chose four classifiers as they performed better in several studies related to IDSs [24, 26–30]. In order to obtain the final feature subsets for univariate filters, three thresholds were used: 20%, 40% and 60%. In fact, based on the previous studies [26, 30, 31], the thresholds 20%, 40% and 60% gave more consistent results over different rankers for different classifiers and datasets. This work also performs parameter tuning using Bayesian Optimization (BO) based on the Bayesian theorem [32] to tune parameters of the four classifiers we used.

Globally, this research compares 228 variants of classifiers: $228 = (4 \text{ classifiers} * 3 \text{ datasets}) * (5 \text{ univariate filters} * 3 \text{ selection thresholds} + 3 \text{ multivariate filters} + \text{original features set})$ using a 10-fold cross-validation method. In addition, this experiment uses: five performance criteria (accuracy, precision, recall, f1-score and area under the ROC curve) to evaluate all trained classifiers, the Scott Knott (SK) test [33] to cluster the classifiers and statistically compare their performance, and the Borda Count (BC) ranking system ranks the highest performing models based on numerous performance criteria.

The following research questions (RQs) were addressed:

1. **RQ1:** Does the dimensionality reduction increase/decrease the performance of ML-based attacks classification?
2. **RQ2:** Do univariate filters perform better than multivariate filters when used to classify attacks on networked systems?
3. **RQ3:** Are there any combinations of FS techniques and classifiers that significantly improve the accuracy of IDSs?
4. **RQ4:** Are there any features that can be considered relevant for attacks classification in all environments and attack scenarios?

The contributions of this empirical evaluation are summarized as follows:

- Improvement of attack detection performance of IDS by evaluating the generalizability of a common feature set to different network environments and attack scenarios.
- Demonstrating that selecting the right set of features for classification can help in enhancing the performance of ML techniques and increasing the chances of potential deployment of IDSs.
- Providing a better combination of ML classifiers and FS methods using statistical analysis such as Scott-Knott test and ranking techniques such as Borda Count.

The paper is organized into the following sections: Sect. 2 briefly describes the different FS techniques used in this study. An overview of the classification and optimization approaches is provided in Sect. 3. Section 4 outlines some related work dealing with the use of FS techniques, especially in IDS. Section 5 describes the entire experimental design. Section 6 shows and discusses the obtained results. Threats to validity are described in Sect. 7. Finally, Sect. 8 presents the conclusions and future works.

2 Feature selection

This section provides an overall view of the FS techniques used in this study. There are two types of methods for reducing the number of features, commonly known as dimensionality reduction: (1) Feature extraction seeks to modify and produce entirely new features based on a combination of original features, for example Principal Component Analysis (PCA) and Linear Discriminant Analysis (LDA); and (2) FS, contrary to feature extraction, selects a subset of features from the original set based on feature redundancy and relevancy to accurately characterize the whole dataset [34].

FS algorithms are means to shorten computation times, gain a deeper knowledge and explainability of the data [35]. It is divided into three major categories [36]: filters, wrappers

and embedded techniques. Filter techniques use statistical measures to select a feature subset based on their association with the target variable, without the need for any training algorithm. Wrappers are iterative techniques that select, at each iteration, a subset of data, train a model and finally, select the best combination based on how well they improve the performance of the trained model [37], while the Embedded models refer to techniques that incorporate FS as part of the model building process itself [38]. Furthermore, filters are faster and require less computational resources than the wrapper methods. Moreover, they are independent of any learner. Filter techniques can be either univariate or multivariate [39]. Univariate techniques, also known as feature ranking, consider only one feature at a time, whereas multivariate techniques consider multiple features at the same time.

2.1 Univariate methods

Univariate filter-based techniques are often used as a preparatory step in the FS process to reveal the most important features and eliminate redundant or irrelevant features from a greater number of features. This study included five univariate filter-based techniques: Chi-squared test, Pearson Correlation, Relief, Mutual Information, and ANOVA test. Each filter is briefly described below.

a) Chi-squared test: A chi-squared test (K) [40] is a statistical approach used to identify features that are dependent on the target variable based on the Chi-2 score.

b) ANOVA test: ANOVA test (A) is an acronym for Analysis of Variance [41]. In this method, each feature is scored using the f-statistic which gives an idea about how far are the means of different classes for each variable. So, a good attribute is the one that best separates the classes.

c) Mutual Information:: Mutual Information (MI) [42] calculates the decrease in entropy of a feature under the condition of the target value. Therefore, a high mutual information value denotes a significant reduction in uncertainty when we consider the feature and the target variable at once, indicating a strong dependence between the two.

d) Relief: The basic principle of Relief (RL) algorithm is to measure the quality of an attribute based on how effectively it can distinguish between instances that are close to one another [43]. A relevant attribute must have the same value for the instance from the nearest hit (nearest neighbor of the same class) and a different value for the instance from the nearest miss (nearest neighbor of a different class).

e) Pearson Correlation: Pearson correlation (C) is a measure of the linear correlation between each feature and the target variable [44].

2.2 Multivariate methods

Multivariate methods are based on the principle of Maximum Relevance Minimum Redundancy (MRMR). It is an iterative algorithm that picks at each iteration the better feature and adds it to the feature subset already chosen [45]. This strategy is known as MRMR because it chooses the feature or attribute with the greatest relevance to the target variable and the least redundancy regarding the attributes selected at earlier iterations. Next subsections briefly describe the three multivariate techniques we used in this empirical evaluation.

a) Correlation-based feature subset selection (CFS): The objective of this technique is to find at each iteration, a new relevant feature that guarantee a high correlation with the target (feature-class correlation), and a low correlation with each variable in the previous subset (feature-feature correlation) at each iteration, a new relevant feature and add it to the subset already chosen [18].

b) Consistency-based feature subset selection (CON): For this technique, the best subset is the one that has the minimal inconsistency rate. A feature subset is said to be inconsistent if there are at least two instances with the same values for all the attributes, but which belong to different classes [46].

c) Double input symmetric relevance (DISR): The DISR [47] method is based on the idea that a set of variables together can yield more information about a target class than the total of the information provided by each feature considered separately.

3 Classification techniques

This section introduces an overview of the classification techniques used in our experiments. There are two single techniques: SVM and MLP; and two ensemble ones: RF and XGB.

a) Support Vector Machine (SVM): This is a supervised learning algorithm for classification and regression tasks that is proposed by Vapnik [48]. The SVM algorithm can generate decision rules with low error on independent test sets, making it effective for solving nonlinear, high-dimensional learning problems with potential for local minima [49, 50]. The two commonly used families of kernels are radial basis function (RBF) and polynomial. There are three main parameters to take into account when using SVM that are [51]: (1) the kernel type, (2) the kernel parameters, and (3) the regularization constant denoted by C .

b) Multilayer Perceptron (MLP): This algorithm is a subclass of feedforward artificial neural networks (ANN) used for classification and regression tasks, and it is viewed as the

fundamental architecture for DL or Deep Neural Networks (DNN) [29, 52]. There are four parameters that must be considered when using the MLP [52]: (1) the number of hidden layers, (2) the learning rate, (3) the number of learning epochs and (4) the momentum.

c) Random Forest (RF): RF is a well-known ensemble learning algorithm made up of many decision trees, which are trained on randomly selected subsets of the data [53]. In addition, there are several hyperparameters that should be taken into consideration when using RF model: (1) number of trees, (2) the maximum depth of each decision tree, (3) the minimum number of samples, (4) the maximum number of features, (5) criterion and (6) bootstrap sampling.

d) XGBoost (XGB): XGBoost (XGB) or (eXtreme Gradient Boosting) is a popular and powerful open-source library for implementing gradient boosting decision trees. The main idea behind gradient boosting is to train a series of weak models, with each model being trained to rectify the mistakes of the prior model [54]. XGB model has several key features that make it more effective than many other gradient boosting libraries, including [55]: (1) the number of boosting estimators, (2) learning rate, (3) subsample ratio of the training features and (4) maximum depth of a tree.

e) Hyperparameters settings: Traditionally, ML techniques rely on model hyperparameters (HP) to adapt their behavior to the dataset to which they are applied [56]. The careful selection HP values has a considerable influence on the performance of ML algorithms. Generally, users must manually specify HP values before starting the learning process, which can be a challenging task as the optimal HP values for any ML technique depend on the unique characteristics of the training dataset. For example, in the case of XGB and MLP models, employing various learning rates is intended to strike a delicate balance between achieving rapid convergence and preventing overfitting. Higher learning rates expedite convergence but heighten the risk of overfitting, while lower rates mitigate overfitting but slow convergence. To address this challenge, this study explores the application of Bayesian optimization (BO) as a strategy to fine-tune the hyperparameters of the four classifiers. Bayesian optimization represents a sophisticated approach that utilizes probabilistic models to search for the best configuration of an ML technique within a specific dataset [32].

4 Related work

FS refers to the technique of identifying the most important and relevant feature subset from a wider set of available features, using specific criteria to guide the selection process [57]. As a result, ML-based IDS can perform predictions

more efficiently. This section provides an overview of some primary studies on the application of FS techniques in different application domains, and in intrusion detection in particular.

Many studies have explored the use of FS techniques for intrusion detection. Kurniabudi et al. [26] used the Information Gain FS technique to select the most significant features, aimed at enhancing the intrusion detection accuracy and reducing the execution time over the CIC-IDS2017 dataset. The results showed that the RF classifier achieved an accuracy of 99.86% by reducing the total number of features from 77 to 22 features. Meanwhile, the J48 reached an accuracy of 99.87% when using 52 selected features. Aksu et al. [27] proposed and evaluated the use of the Fisher Score FS algorithm and three classifiers: SVM, KNN, and DT to classify data flows as either DDoS or benign over the CIC-IDS2017 dataset. The SVM classifier outperformed the others and reached an accuracy of 99.97%. According to Dhaliwal et al. [30], the ML competition site "Kaggle" shows that most winning solutions use XGB (17 out of 29 winning solutions used XGB in 2015). Ghosh et al. [31] presented a hybrid FS approach merging the filter and the wrapper methods for anomaly-based IDS. The experiment decreased the number of features from 79 to 11, which is almost 20% of the entire feature set, resulting in a classification accuracy of 99.82%. Uzun et al. [58] proposed a new approach to increase the effectiveness of IDSs, by utilizing a combination of multivariate outlier detection and optimal FS over the NSL-KDD dataset. The results showed that the Relieff technique yielded statistically significant results with quick detection speed, and the accuracy reached a value of 99.21% with only 20 relevant features among 41 features from the original feature set. An ensemble-based IDS model was proposed by Nutan et al. [59] using an integrated FS approach. This paper offered some intrusion detection methodologies based on ML methods (Bayesian Network, J48 and Naive Bayes) with wrapper approaches. The findings showed that the features were reduced from 41 to 12, and the overall accuracy was 98.46%. In [60], Zhou et al. developed and evaluated a new intrusion detection method using an ensemble based on FS over the CIC-IDS2017, NSL-KDD and AWID datasets. The work introduced the CFS-BA heuristic approach for dimensionality reduction, using an ensemble approach combining RF, C4.5, and Forest by Penalizing Attributes (Forest PA) algorithms. The results demonstrated that the use of CFS-BA-Ensemble FS technique improved the attacks classification performance which reached a classification accuracy of 99.89% using a subset of 13 features out of 78 ones from the original feature set of the CIC-IDS2017 dataset, 99.81% with a subset of 10 features out of 41 features from the original feature set of the NSL-KDD dataset, and 99.52% reducing the total number of features from 83 to 8 features for the AWID dataset. Linhares et al. [61] presented a

SDNTruth system, a mechanism for mitigating DDoS attacks in software-defined networks (SDN) using statistical analysis and traffic entropy. In this study, the authors conducted this approach by using only four features and demonstrated that statistical techniques can produce good results in detecting DDoS attacks in SDN with much less consumption of computational resources.

Table 1 resumes the findings of the previous selected papers that specifically investigate the use of FS techniques for IDSs.

5 Experimental design

This section briefly describes the datasets used, the performance metrics and statistical tests applied to evaluate and compare the models constructed, and the methodology followed.

5.1 Datasets description

Figure 1 illustrates the procedure of traffic flow generation using CICFlowMeter tool. The procedure starts with extracting a large range of attributes from network flow data (in pcap format), and then labeling the extracted flow records in the final step (in csv format). Once the attributes are extracted, they can be used for further analysis or to train ML models.

The present study uses three datasets related to intrusion detection, which are characterized by a common set of features.

- **CIC-IDS2017** [62]: This is a dataset of network traffic for intrusion detection, collected by the Canadian Institute for Cybersecurity [63]. It was collected from network traffic in an isolated environment and simulated the actions of 25 legitimate users based on different protocols such as HTTP, HTTPS, SSH, FTP and email. The malicious traffic includes a variety of different types of attacks, such as Infiltration, Brute Force, DoS, Distributed DoS (DDoS), Heartbleed, Web Attack and Botnet. This dataset is spread across 8 csv files, holding five days of normal and intrusion traffic, which have a total number of 79 features extracted by the CICFlowMeter tool on the basis of the information present in the pcap file. Statistically, the whole number of instances is 2,522,362, of which 2,096,484 (83%) are benign and 425,878 (17%) are attack samples, presenting a disequilibrium between the classes as shown in Fig. 2a.
- **CSE-CIC-IDS2018** [64]: This project is a joint effort between the Canadian Institute for Cybersecurity (CIC) and the Communications Security Establishment (CSE). It covers seven attack scenarios (Infiltration, Brute-force, Botnet, Web attacks, DoS, DDoS, and Heartbleed). The

Table 1 Summary of the literature review

Paper	Dataset	Classifier	FS techniques	# Of selected features	Best accuracy	Finding
Our study	CIC-IDS2017, CSE-CIC-IDS2018 and CIC-ToN-IoT	SVM, MLP, RF and XGB	Relieff, Pearson Correlation, Mutual Information, ANOVA, Chi2, CFS, CSS and DISR	13 (CIC-IDS2017), 6 (CIC-IDS2018), 16 (CIC-ToN-IoT)	99.92% (CIC-IDS2017), 99.99% (CSE-IDS2018) and 99.22% (CIC-ToN-IoT)	Evaluate and compare the impacts of univariate and multivariate filter on the performances of four classifiers in IDS. The findings show the potential of the use XGB and RF models with CON and DSIR techniques since they give a very robust performances in terms of accuracy and calculation time, even when using a small number of features
Uzun et al. [58]	NSL-KDD	RF	Relieff	20	99.21%	Propose a novel approach based on RL technique to discover the best features that keep classification performance high, and 20 features were determined. With 99.21% accuracy, the RF model has achieved the best classification success
Nutan et al. [59]	NSL-KDD	Bayesian Network, Naive Bayes, J48 and ensemble approach based on majority vote	Wrapper approach: Best first, genetic search and rank search	12	98.46%	Propose an approach of classification model that integrates the Naive Bayes, Bayesian Network, and J48 algorithms with a wrapper-based hybrid FS technique. The use of 12 essential features enables an accepted FPR of 0.021. The findings demonstrate that the proposed approach outperforms the Naive Bayes, J48 and Bayesian Network classifiers

Table 1 continued

Paper	Dataset	Classifier	FS techniques	# Of selected features	Best accuracy	Finding
Zhou et al. [60]	AWID, NSL-KDD, CIC-IDS2017	C4.5, RF and Forest by Penalizing Attributes (Forest PA) are combined in an ensemble technique	CFS	10 (NSL-KDD), 8(AWID) and 13 (CIC-IDS2017)	99.81% (NSL-KDD), 99.52% (AWID) and 99.89% (CIC-IDS2017)	Develop a framework for IDS based on the FS and an ensemble learning algorithm. The authors employed a heuristic technique called CFS-BA to select the best subset based on feature correlation, and then constructed an ensemble approach consisting of voting three models. The findings show the effectiveness of the CFS-BA Ensemble method, which obtained accuracy of 99.81%, 99.52%, and 99.89% on the NSL-KDD, AWID, and CIC-IDS2017 datasets, respectively
Kumiabudi et al. [26]	CIC-IDS2017	Bayes Net, Random Tree, Naive Bayes, J48 and RF	Information gain	22	99.86%	Reveal the important and significant features using Information Gain technique, the authors constructed five algorithms on CICIDS-2017 dataset. The findings show that FS has the ability to enhance detection accuracy and execution time. With 22 relevant features, RF model achieved an accuracy of 99.86%

Table 1 continued

Paper	Dataset	Classifier	FS techniques	# Of selected features	Best accuracy	Finding
Aksu et al. [27]	CIC-IDS2017	SVM, K-Nearest-Neighbour and decision tree	Fisher score	30	99.97%	Analyze the effects of FS techniques in the performances of three models: SVM, KNN and decision tree. The authors selected the top features from the CIC-IDS2017 dataset using the Fisher Score approach. The results show that RF outperformed all the single models using 30 relevant features, with a 99.97% accuracy rate
Ghosh et al. [31]	CSE-CIC-IDS2018	SVM, K-Nearest-Neighbour and decision tree	The filter and wrapper procedures are combined in a hybrid FS approach	11	99.82%	Propose a hybrid FS techniques designed using filter and wrapper approaches. The authors evaluated on the DDOS attack of the CSE-CIC-IDS2018 dataset, and the results show that by using the 11 most important features, the accuracy achieved 99.82%
Linhares et al. [61]	SimpleWeb site	DT, RF, ANN, LR, SVM and NB	–	04	97.41%	Evaluate and compare the viability and effectiveness of a new approach based on statistical analysis and entropy in SDN. The results show the potential of using statistical techniques since they give very robust performance in terms of accuracy and computation time, even when using a small number of features, the accuracy achieved 97.41%

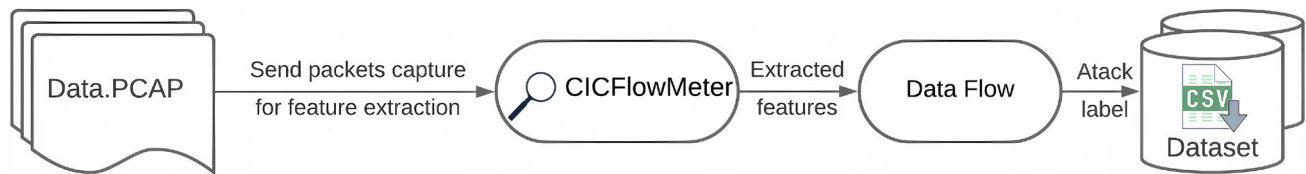


Fig. 1 Features extraction process

attacker's infrastructure consists of 50 machines, while the victim organization consists of 5 departments, each with 420 machines and 30 servers. The dataset contains network traffic captures and system logs from each machine, as well as 84 features extracted from the traffic gathered with the CICFlowMeter tool. This dataset contains about 16,252,172 records collected over a period of 10 days available in ten different (csv) files. In addition, there are 2,352,951 (15%) intrusion samples and

13,352,501 (85%) normal samples. Additionally, it contains a variety of attacks, as seen in Fig. 2b, but the focus is on two types of attacks DDoS, and Bot attacks.

- CIC-ToN-IoT:** This dataset was produced as part of the study of Sarhan and al. [65], wherein the feature set was extracted from the ToN-IoT dataset [66] using CICFlowMeter tool. The ToN-IoT dataset was created in parallel processing utilizing an industrial network testbed and numerous virtual computers running Windows, Linux, and Kali Linux operating system hosts, and it was used to capture multiple normal and cyber-attack events from IoT networks [67]. It includes Backdoor, DoS, DDoS, Injection, Man In The Middle (MITM), Password, Ransomware, Scanning, and Cross-Site Scripting (XSS) attack scenarios. As shown in Fig. 2c, the CIC-ToN-IoT dataset contains a total of 5,351,760 data samples, of which 2,836,524 (53.00%) are intrusion samples and 2,515,236 (47.00%) are normal samples.

A short description of all features designed by the CICFlowMeter is presented in Table 15 of Appendix A.

5.2 Performance measures and statistical tests

- Methods for Model Evaluation and Validation: Performance measures are used to evaluate and compare the results of ML models. Many metrics can be utilized in classification tasks that depend on a confusion matrix. Table 2 gives the elements of a 2 * 2 confusion matrix in which the diagonal elements represent the correct classification (The values of True-Positive (TP) and True-Negative (TN)). In this experiment, the following metrics were considered:

-TP (True Positive): All cases with positive actual value and positive predictive value.

-FP (False Positive): All cases with actual negative value and positive predictive value.

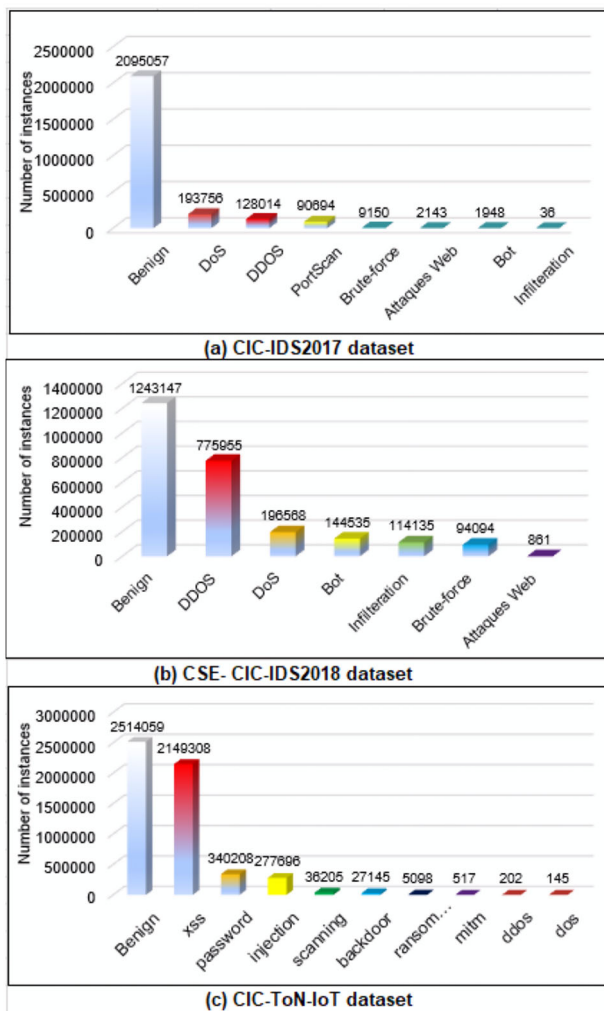


Fig. 2 Distribution of attack types of each dataset. The x-axis illustrates the different class attack scenarios. The y-axis illustrates the number of instances by class

Table 2 Confusion matrix

	Predicted intrusion	Predicted normal
Actual intrusion	True positive (TP)	False negative (FN)
Actual normal	False positive (FP)	True negative (TN)

-FN (False Negative): All cases with actual positive value and negative predictive value.

-TN (True Negative): All cases with actual negative value and negative predictive value.

The following metrics are used to evaluate the performance of all variants of classifiers.

Accuracy (ACC), defined in Eq. 1, represents the number of correct predictions made by a model, divided by the total number of the dataset samples.

Precision (PREC), defined in Eq. 2, quantifies the reliability of a model's positive predictions, obtained by dividing the number of correct positive predictions by the total number of correct positive predictions made by the model.

F1-score, defined in Eq. 3, is a metric that combines the precision and recall metrics to assess a model's accuracy on a dataset. It is defined as the harmonic mean of precision and recall.

Recall, defined in Eq. 4, also known as sensitivity or TPR, represents the number of correct positive predictions made by a model, divided by the total number of positives.

AUC gives the area under the ROC curve (Receiver Operating Characteristic), it is a performance metric for classification problems. It represents the separability of classes, with a higher AUC indicating better distinction between malicious and non-malicious packets. The AUC is derived from the ROC curve, which shows the model's accuracy at different thresholds.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$F1 - \text{score} = \frac{2TP}{2TP + FN + FP} \quad (3)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

Statistical analysis: For the statistical analysis, Scott-Knott (SK) test was used to compare the performances of the classifiers and to determine the significance of the results obtained from the model's evaluation. SK test is a statistical method proposed in 1974 by Scott and Knott [68]. The SK test is a hierarchical clustering method applied as a tool for exploratory data analysis to partition treatments into distinct groups. It was proposed to help researchers doing different comparison of treatment means, to find distinct homogeneous groups of these means [33].

Ranking technique: Borda Count Voting System (BC) is a voting technique where voters rank candidates in order of preference [69]. Candidates are awarded points based on their ranking, and the one with the most points wins. In ML, models are considered candidates and performance metrics are considered voters.

5.3 Methodology

The methodology applied in this empirical study stands out for its holistic approach to IDSs evaluation compared with the majority of existing work. Unlike some studies [26, 27, 31, 58–60] that limit their examination to a restricted number of feature selection techniques and a selected set of classifiers, this research adopts a broader perspective. The study systematically evaluates the impact of numerous feature selection methods, including a unique comparison of five univariate filters with three multivariate filters. This methodology is reinforced by the use of a varied set of evaluation criteria and advanced statistical techniques, guaranteeing the credibility and reliability of the results.

Figure 3 describes the overall process followed to evaluate and compare the impact of FS on the classification performance in IDS over three datasets CIC-ID2017, CSE-CIC-ID2018, and CIC-ToN-IoT. The four classifiers' (SVM, MLP, RF and XGB) performances were evaluated using a 10-fold cross-validation approach.

The experiment process consists of:

Step1: Transform the raw data into a usable and correct format by cleaning each dataset of missing values, duplicate rows, and irrelevant attributes. Thereafter, we apply the eight univariate/multivariate filters to each dataset:

Univariate techniques (A, K, C, MI and RL) return a list of ranked features: A ranking list of features is returned by applying each univariate technique, and three thresholds (20%, 40%, and 60%) were used to select the top-ranked features. These thresholds were chosen since they covered the most frequently used percentages: Kurniabudi et al. [26] considered 22 (thresholds of 67%) out of 77 as the best features when using an RF classifier, and 52 (thresholds of 29%) features using j48 classifier. Ghosh et al. [31] conducted an experiment for reducing the total number of features from 79 to 11 features (thresholds of 14%), resulting in the classification accuracy of 99.82%. Aksu et al. [27] reduced the original feature set from 80 to 30 (thresholds of 38%) using SVM algorithm and achieved an accuracy of 99.97%. Jairu and al. [28] demonstrated that RF outperforms other supervised algorithms using only 14 (thresholds of 19%) features out of 77 features from the original feature set. Other results of an experiment conducted by Fitni et al. [70] showed that, among the 80 features, 23 (thresholds of 28%) were selected using Spearman's rank correlation coefficient and achieved 98.8% accuracy. Uzun et al. [58] showed that the RF model has achieved a high classification performance 99.2187%, with only 20 relevant features out of 41 features from the original set of features (thresholds of 48%). Nutan et al. [59] demonstrated that the features were reduced from 41 to 12 (thresholds of 30%), and the overall accuracy was 98.46%. In [60], Zhou and al, reduced the original feature set from three datasets: For the CIC-IDS2017 dataset, the original

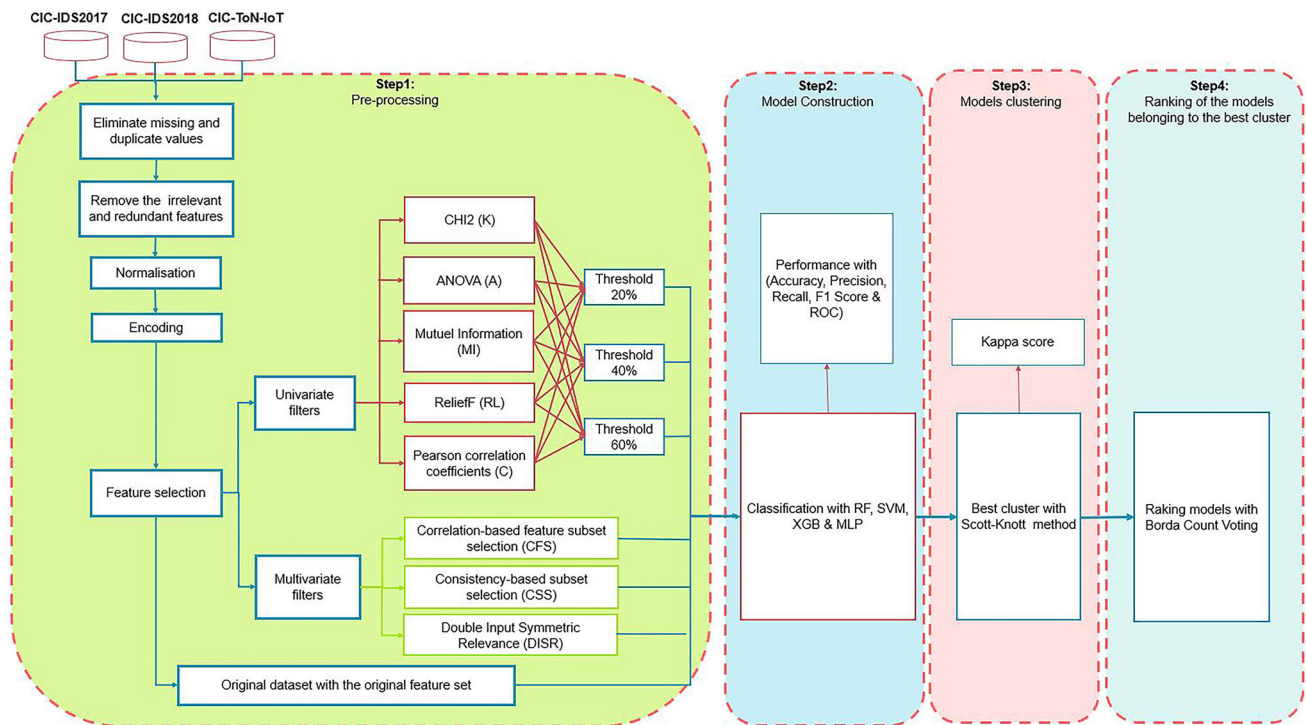


Fig. 3 Overall process of the proposed FS-based IDS

feature set was reduced from 78 to 12 (thresholds of 16%) features and the classification performance achieved an accuracy of 99.97%. For NSL-KDD dataset, the original feature set was reduced from 41 to 10 (thresholds of 24%) features and the accuracy achieved 99.97%. For AWID dataset, the original feature set was decreased from 83 to 8 (thresholds of 10%) features and the accuracy was 99.97%. In conclusion, thresholds of 20%, 40%, and 60% gave more consistent results across different FS techniques and datasets. In total, 45 feature subsets were obtained (Five univariate techniques applied to three datasets with three percentages).

Multivariate techniques (CFS, CSS and DISR) return a feature subset that is the most relevant to the target variable. These multivariate filters were frequently investigated when dealing with FS in IDS [71, 72]. By applying these three techniques to the three datasets, a total of 9 subsets of features were obtained.

Step2: Construct and evaluate the performances of the four classifiers SVM, MLP, RF, and XGB using a 10-fold cross-validation over each dataset of Step1, as well as over each original dataset (i.e., included all the features). The pre-defined search spaces of the Bayesian Optimization for each classification technique are listed in Table 3. For each dataset, 19 subsets of features (5 univariate filters * 3 thresholds) + 3 multivariate filters + the original feature set) were used by each of the four classifiers (SVM, MLP, RF, XGB). In total, we obtain 19 variants of each classifier per dataset.

Step3: Use the SK test to cluster the classifiers on the basis of accuracy criterion to statistically test the significance of classifiers performance differences.

Step4: Rank the classifiers belonging to the best SK cluster using the BC system, based on the five metrics: Accuracy, F1 score, Precision, Recall and AUC. In order to facilitate the naming of the constructed models, the percentages 20, 40 and 60 are added next to model abbreviations to indicate the threshold used for the selected feature subset. For example, SVMRL40 classifier refers to SVM model trained with the feature subset selected by Relieff with the 40% threshold. Additionally, for the original datasets with all features, the number 100 indicates the original feature set for each dataset. For example, SVM100 refers to the variant of SVM which used the whole features of each dataset.

6 Results and discussion

This study performed a comparative analysis of four classifiers: SVM, MLP, RF and XGB that were trained/evaluated with different feature subsets selected from three datasets related to network and IoT data. The feature subsets were selected using five univariate and three multivariate FS techniques. The results of this empirical evaluation are provided and discussed in this section in terms of the RQs presented in Sect. 1.

Table 3 Search spaces of HPs values used by Bayesian optimization for the four classification techniques

Classifier	Hyperparameters	Values
SVM	Kernel	[poly, rbf, sigmoid]
	C	[1, 200] with an increment of 10
MLP	Gamma RBF	[0.001, 0.01, 0.05, 0.1]
	Alpha	[0.000001, 0.00001, 0.0001, 0.001]
	Hidden layer sizes	[1, 30] with an increment of 1
	Learning rate	constant, invscaling, adaptive
	Solver	lbfgs, adam, SGD
	Activation	relu
RF	Momentum	[0.001, 0.01, 0.05, 0.1]
	N estimators	[200, 1400] with an increment of 200
	Max-features	[auto, log2, sqrt]
	Max-depth	[4, 28] with an increment of 4
	Min samples split	[1, 5] with an increment of 1
	Min samples leaf	[1, 5] with an increment of 1
	Bootstrap	[True, False]
	Min child weight	[1, 40] with an increment of 5
XGB	Gamma	0.5 and [1, 100] with an increment of 10
	Subsample	[0.2, 1] with an increment of 0.2
	Colsample bytree	[0.2, 1] with an increment of 0.2
	Max depth	[2, 20] with an increment of 2
	Learning rate	[0.0001, 0.001, 0.01, 0.05, 0.1, 0.2]

6.1 Data pre-processing

The pre-processing shown in Fig. 4 consists of stages that must be taken to ensure that the datasets are in an appropriate format before being processed by our classifiers.

- For *CIC-IDS2017 dataset*: 1576 instances of infinite values and a total of 51,902 duplicated instances were removed. In addition, 20 features with a correlation value greater than 95% were removed using the correlation coefficient [73]. Moreover, 8 constants and 8 duplicated features were also removed.

- For *CIC-IDS2018 dataset*: A total of 94,296 instances with partial or missing information were deleted, and a list of 38 attributes were removed since they are constant, duplicated, or highly correlated attributes.
- For *CIC-ToN-IoT dataset*: 1177 instances with missing value were deleted, and a total of 33 irrelevant attributes were dropped from the dataset because they were a combination of constants, highly correlated or duplicated features. The remaining features were then encoded and normalized to ensure that the dataset was ready for training.

By keeping all data in the minority class and lowering the size of the majority class, the under-sampling strategy was used in this study to balance the distribution of classes. Figure 4 presents the number of samples of each class after handling all imbalanced datasets.

Table 4 summarizes the number of features with their types, the number of instances, and classes for each preprocessed dataset.

Note that all the feature subsets obtained by means of univariate or multivariate filters as well as those preprocessed by the authors to handle infinite values and irrelevant features are available upon request by email to the corresponding author of this paper.

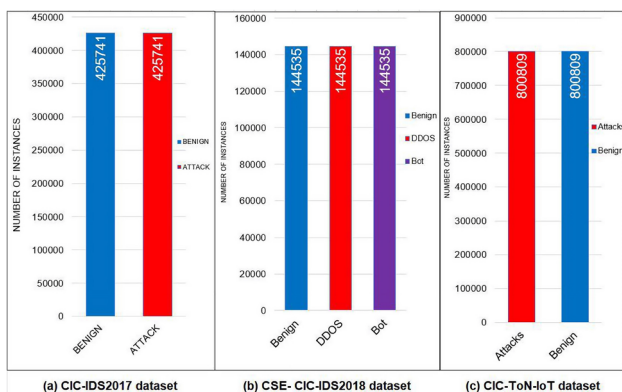


Fig. 4 Number of samples of each class after balancing datasets

Table 4 Datasets description after the preprocessing step

Dataset	# Of informative features	# Of instances	# Of categorical features	# Of numerical features	# Of classes
CIC-IDS2017	42	851,619	8	34	2
CSE-CIC-IDS2018	44	433,605	10	34	3
CIC-ToN-IoT	50	1,601,618	3	47	2

6.2 Impact of dimensionality reduction in the performance of attacks classification (RQ1)

This subsection presents the results of the step3 of our methodology shown in Fig. 3. For each dataset, a total of 18 feature subsets (5 univariate filters * 3 thresholds + 3 multivariate filters) were used to assess the performances of the four models (SVM, MLP, XGB and RF). Additionally, we assess how well the four models (SVM, MLP, XGB, and RF) perform using the original datasets (i.e., without using FS). This leads to 19 variants of each couple (classifier, dataset) that were evaluated to identify the best cluster using the SK test based on accuracy. Figures 5, 6 and 7 show the SK results of each couple (classifier, dataset). In total, we obtain 12 SK figures (4 classifiers * 3 datasets).

SK results are represented by two axes, the x -axis represents the variants of the classifier sorted by cluster, where the best clusters names start from left to right, and the y -axis represents the accuracy scores. The little dots in the center of each vertical line reflect the average accuracy, and each line

depicts the fivefold cross-validation results for a classifier. From Figs. 5, 6 and 7, it can be observed that all classifiers trained with the original dataset were generally positioned in the first or second cluster with the exception of the SVM100 over the CIC-IDS2017 dataset.

To determine how the FS techniques affect the effectiveness of the classifiers employed, a statistical analysis based on the SK test was performed (Figs. 5, 6 and 7). Table 5 lists the number of classifiers positioned in the same cluster of the classifier trained with the original feature set (given by the column S: cases in where FS has no impact), the number of classifiers positioned in the left side of the classifier trained with the original feature set (given by the column W: cases in where FS has a positive impact) and the number of classifiers positioned in the right side of the classifier trained with the original feature set (given by the column L: cases in where FS has a negative impact). If W (L) is the highest, then FS techniques positively (negatively) impact the classifiers performance. A high value of S indicates that FS techniques have no impact on the classification performance. Addition-

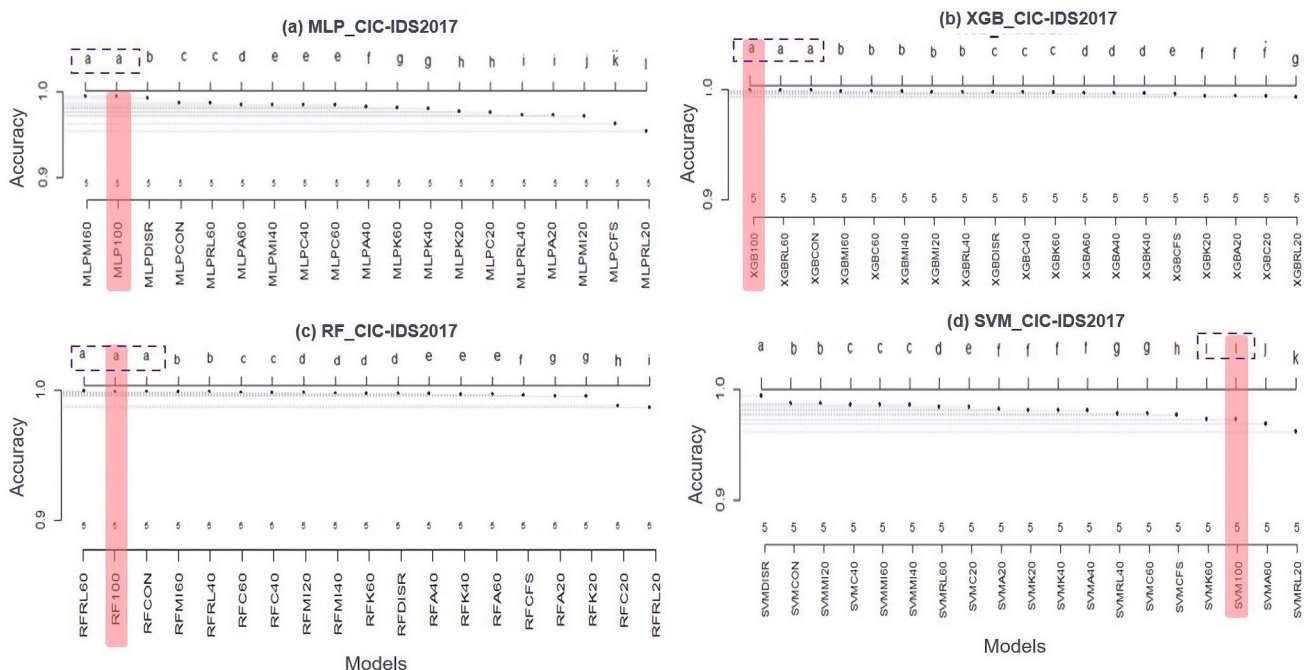


Fig. 5 SK test results for CIC-IDS2017 dataset by model. The generated classifiers are represented by the x -axis, and accuracy scores are represented by the y -axis. The best clusters ranking start from the left side to the right side

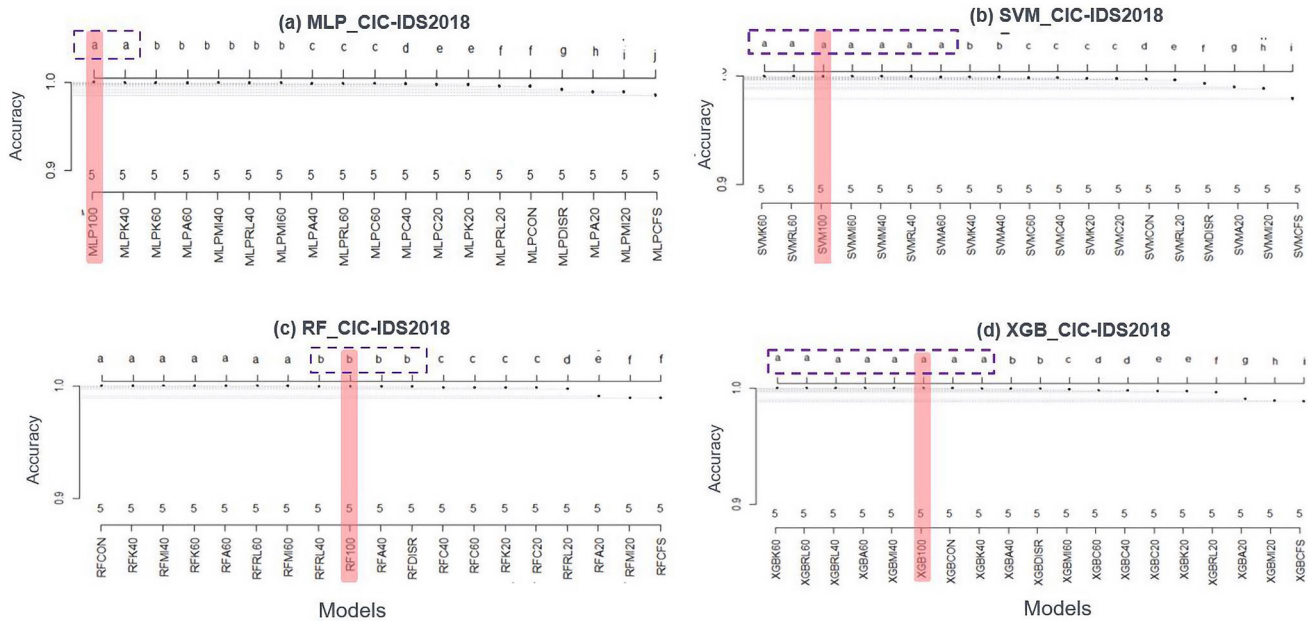


Fig. 6 SK test results for CSE-CIC-IDS2018 dataset by model. The generated classifiers are represented by the x-axis, and accuracy scores are represented by the y-axis. The best clusters ranking start from the left side to the right side

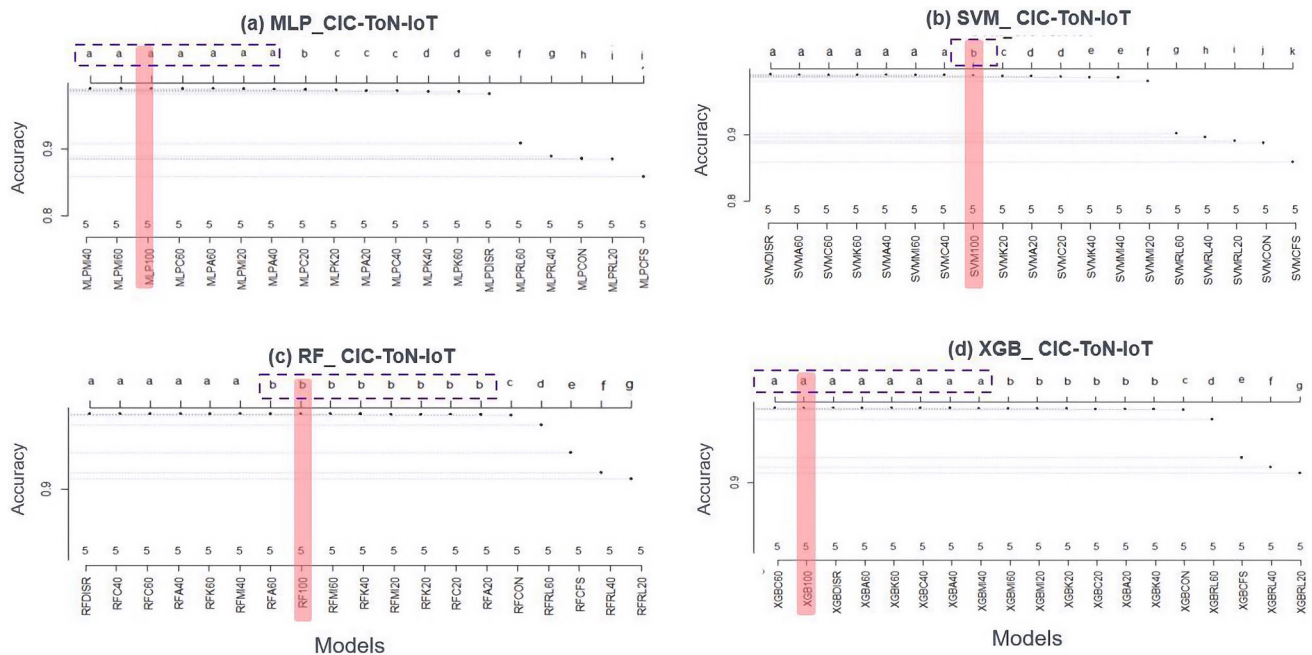


Fig. 7 SK test results for CIC-ToN-IoT dataset by model. The generated classifiers are represented by the x-axis, and accuracy scores are represented by the y-axis. The best clusters ranking start from the left side to the right side

ally, for each model in each dataset in Table 5, all maximum values among S, L and W are written in bold.

For instance, for CSE-CIC-IDS2018 dataset, the RF100 classifier in Fig. 6c is positioned in the second cluster, the number of classifiers in the same cluster is 3 ($S=3$), the number of classifiers positioned in the first cluster in the left is 7

($W=7$), and the number of classifiers positioned in the right of RF100 is eight ($L=8$).

Table 5 Impact of FS on the models' performance based on the SK test results considering all clusters

Model	CIC-IDS2017			CSE-CIC-IDS2018			CIC-ToN-IoT		
	L	S	W	L	S	W	L	S	W
MLP	17	1	0	17	1	0	12	6	0
SVM	2	1	15	12	6	0	11	0	7
RF	16	2	0	8	3	7	5	7	6
XGB	16	2	0	11	7	0	11	7	0
TOTAL	50	7	15	48	17	7	39	20	13

Table 6 Impact of FS on the models' performance across all datasets considering all clusters

Model	L	S	W
MLP	46	8	0
SVM	24	8	22
RF	29	12	13
XGB	38	16	0
TOTAL	137	44	35

From Table 5, it can be noticed that:

- For CIC-IDS2017 dataset, we noticed that, except for SVM model the FS techniques have a negative impact because the value of L is the highest for all models.
- For CSE-CIC-IDS2018 dataset, it can be observed that for all models the FS techniques have a negative impact since the value of L is the highest.
- For CIC-ToN-IoT dataset, we noticed that for RF model the impact was the same because the value of S is the highest. For MLP, SVM and XGB models, the FS techniques have a negative since the value of S is the greatest.

Table 6 summarizes the results of this analysis across all datasets, where values in bold indicate the highest values. It can be observed that the value of L is higher (137), which can indicate that the impact of FS techniques negatively impacted the classification performance. This may be due to the large number of filters used in this study. This allows us to make some observations: (1) the choice of FS techniques must be well done to keep at least the same performance of the classifier that was trained on the original feature set, and (2) A set of recommendations should be established to guide selecting the best combination of FS and classifier.

In order to confirm that choosing the top combinations of FS techniques and classifiers can lead to improve the classification performance, we restrict our analysis to the two best SK clusters. For instance, for CSE-CIC-IDS2018 dataset, the RF100 classifier in Fig. 6c is positioned in the second cluster, the number of classifiers in the same cluster is 3 (S=3), the number of classifiers positioned in the first cluster in the left is 7 (W=7), and the number of classifiers positioned in the right of RF100 is zero (L=0) for the two best clusters.

Table 7 Impact of FS on the models' performance based on the SK test results considering the two best clusters

Model	CIC-IDS2017			CSE-CIC-IDS2018			CIC-ToN-IoT		
	L	S	W	L	S	W	L	S	W
MLP	1	1	0	6	1	0	1	6	0
SVM	0	0	3	2	6	0	0	0	7
RF	1	3	0	0	3	7	0	7	6
XGB	5	3	0	2	7	0	6	7	0
TOTAL	7	7	3	10	17	7	7	20	13

Table 8 Impact of FS on the models' performance across all datasets considering the two best clusters

Model	L	S	W
MLP	8	8	0
SVM	2	6	10
RF	1	13	13
XGB	13	17	0
TOTAL	24	44	23

Furthermore, all maximum values in Table 7 among S, L and W are written in bold. When considering each dataset separately, it can noticed that:

- For CIC-IDS2017 dataset, we noticed that for SVM, the FS techniques have a positive impact because the number of W is the highest. For MLP and RF models, the impact was the same because the number of S is the greatest. For XGB, the impact of FS techniques is negative since the number of L is the highest.
- For CSE-CIC-IDS2018 dataset, we observed that for RF model the FS techniques have a positive impact because the number of W is the highest. For SVM and XGB models, the impact was the same because the number of S is the greatest. For MLP, the impact of FS techniques is negative since the number of L is the highest.
- For CIC-ToN-IoT dataset, we noticed that for SVM model the FS techniques have a positive impact because the number of W is the highest. For MLP, RF and XGB models, the impact was the same because the number of S is the greatest.

In sum, Table 8 summarizes the results of this analysis across all datasets, where the highest values are written in bold. It can be noticed that the dimensionality reduction increases the performance of SVM and RF models in most cases, which is not the case of the MLP and XGB models. Furthermore, we observe that the impact of the FS techniques across all datasets is the same, since the value of S (44) is the highest.

In general, the application of FS techniques has a positive impact if the FS-Classifier combinations are well chosen,

as it may reduce computation time and models complexity by improving their interpretability while maintaining high performance.

6.3 Comparison of univariate and multivariate filters (RQ2)

This section shows and discusses the number of occurrences of each FS techniques in the three best clusters according to the results of SK test. For every dataset, 76 classifiers (variants of SVM, MLP, RF and XGB using univariate and multivariate filters) were evaluated to identify the best cluster over each dataset. In Fig. 8, which is represented by two axes, the accuracy results of the SK test for all datasets are shown. The x-axis shows the classifier variations, with the better placements starting on the left, while the y-axis shows the accuracy values. The vertical line depicts the fivefold cross-validation results for one variant and each little dots in the center of each vertical line indicate the average accuracy values.

From Fig. 8a, it can be observed that the SK test over the CIC-IDS2017 dataset identified 27 clusters. The first cluster contains 6 variants of classifiers, the second cluster contains 7 variants of classifiers, and the third one contains 10 variants of classifiers.

Figure 8b shows that the SK test over the CSE-CIC-IDS2018 dataset identified 18 clusters. As it can be observed the first cluster contains 15 variants of classifiers, the second cluster contains 7 variants of classifiers and the third one contains 13 variants of classifiers. As for Fig. 8c, it shows that a total of 27 SK clusters over the CIC-ToN-IoT dataset were identified. Out of 76 variants, the first/second/third SK cluster contains 14/13/7 variants of classifiers, respectively.

Table 9 shows the total of univariate and multivariate FS techniques present in the three best clusters of each dataset. Taking into consideration the initial number of univariate (60 filters) and multivariate (12 filters) techniques used over all datasets, 40.5% of univariate and 30.5% of multivariate FS techniques were present in the three best SK clusters over all datasets.

Moreover, taking into account each dataset separately:

- For CIC-IDS2017 dataset, 28.33% of FS techniques present in the three best SK cluster are univariate, while 33.33% are multivariate.
- For CSE-CIC-IDS2018 dataset, 46.66% of FS techniques present in the three best SK cluster are univariate, while 33.33% are multivariate.

Table 9 The occurrence rate of classifiers in the 3 best cluster based on FS techniques for each dataset

	CIC-IDS2017		CSE-CIC-IDS2018		CIC-ToN-IoT		All datasets	
	% Of classifiers (%)	# Of occurrences	% Of classifiers (%)	# Of occurrences	% Of classifiers (%)	# Of occurrences	% Of classifiers (%)	# Of occurrences
Univariate filters (60)								
First cluster	3.33	2	18.33	11	18.33	11	13.33	24
Second cluster	11.66	7	10	6	20	12	13.88	25
Third cluster	13.33	8	18.33	11	8.33	5	13.33	24
Total	28.33	17	46.66	28	46.66	28	40.5	73
Multivariate filters (12)								
First cluster	16.66	2	16.66	2	16.66	2	16.66	6
Second cluster	0	0	0	0	0	0	0	0
Third cluster	16.66	2	16.66	2	8.33	1	13.88	5
Total	33.33	4	33.33	4	25	3	30.5	11

The values in bold indicate the total percentage in each case

Table 11 Borda count results for CSE-CIC-IDS2018 dataset

Classifier	# Of feature in the subset	Accuracy%	Rank
XGBK60	26	99.99	1
XGBRL60	26	99.99	1
RFCON	6	99.99	2
RFK40	18	99.99	3
XGBRL40	18	99.99	4
XGBMI40	18	99.99	5
RFMI40	18	99.99	6
XGBA60	26	99.99	7
MLP100	44	99.99	8
RFK60	26	99.99	9
RFA60	26	99.98	10
RFRL60	26	99.98	11
XGBCON	6	99.98	12
RFMI60	26	99.98	13
XGB100	44	99.98	14

Table 12 Borda count results for CIC-ToN-IoT dataset

Classifier	# Of feature in the subset	Accuracy%	Rank
XGBC60	30	99.23	1
XGBDISR	16	99.22	2
XGB100	50	99.22	3
RFDISR	16	99.22	4
XGBK60	30	99.22	5
XGBC40	20	99.21	6
XGBA60	30	99.20	7
XGBA40	20	99.20	8
XGBMI40	20	99.19	9
RFC40	20	99.18	10
RFA40	20	99.17	11
RFK60	30	99.17	12
RFC60	30	99.17	13
RFMI40	20	99.16	14

vided by K60 and RL60, respectively, are ranked first with an accuracy of 99.99% for each. We found that **RFCON** and **XGBCON** took 3rd and 12th place, respectively, and trained with the less number of features (**6 features**), with corresponding accuracy rates of 99.99% and 99.98%. Table 12 shows the BC ranks of the 14 classifiers from the best cluster in the CIC-ToN-IoT dataset. The XGB60 trained with 30 features using the C60 was the first ranked with an accuracy of 99.23%. We noticed that the two classifiers **XGBDISR** and **RFDISR** trained with the smaller number of features (**16 features**) were actually ranked second and fourth, with accuracy of 99.22% both.

Table 13 Ranks of the optimal classifier in each dataset

	CIC-IDS2017	CSE-CIC-IDS2018	CIC-ToN-IoT
Optimal models	Cluster number		
RFCON	1	1	4
XGBCON	1	1	4
RFDISR	3	3	1
XGBDISR	3	3	1

For Classifiers: As shown in Tables 10, 11 and 12, the XGB and RF classifiers appear in all best clusters, while the MLP and SVM classifiers have been absent in all best clusters that means XGB, and RF classifiers proved to be very powerful on all datasets.

For FS techniques: From Table 13, it can be shown that, CON technique appears in the first cluster for CIC-IDS2017 and CIC-IDS2018 datasets and in the fourth one for CIC-ToN-IoT dataset. For DISR technique, it appears in the third cluster for CIC-IDS2017 and CIC-IDS2018 datasets and the first one for CIC-ToN-IoT dataset. Furthermore, XGBCON and RFCON combinations may be beneficial to network traffic for intrusion detection, and XGBDISR and RFDISR combinations appear to be more effective for the IoT data set.

The results indicate that certain models, such as XGBs and RFs classifiers, have performed well across all datasets used, this is due to the fact that they are ensemble models. RF and XGB, when paired with univariate filters, exhibit robustness and high accuracy. For the MLP and SVM models, they are not suitable for FS because they are not in the best clusters, with the exception of the MLP100 classifier of CSE-CICIDS2018 dataset. However, CON and DSIR multivariate filters may be slightly less efficient compared to the best classifiers, but offer advantages such as faster learning, a simpler and more interpretable model since the number of features is less important. Finally, from Tables 10, 11 and 12, the best combination of classifiers and FS techniques for each dataset have been written in bold, and across all datasets, it is RF and XGB with CON techniques.

6.5 Relevant features for attacks classification (RQ4)

This subsection presents the number of features resulting from applying the different univariate and the multivariate filters and discuss the most important features for attacks classification in all environments and attack scenarios. For each dataset, a total of 18 subsets were obtained by applying the FS methods. Table 16 and 17 of Appendix A summarize the number of features resulting from applying the univariate and multivariate FS techniques on the three datasets with 20%, 40% and 60% thresholds. It shows that CFS is the technique that produces the minimum number of features

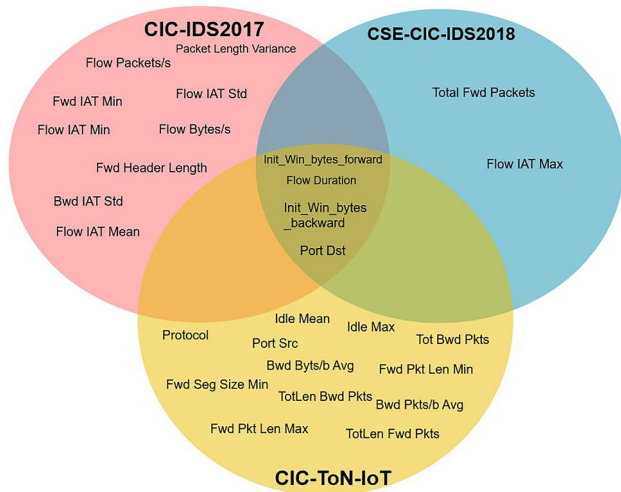


Fig. 9 Feature subsets used to train the optimal classifiers

comparing to the other techniques. Moreover, a total of 8, 17 and 25 features were selected with the thresholds 20%, 40% and 60%, respectively, in CIC-IDS2017 dataset. For CSE-CIC-IDS2018 dataset, the 20%, 40% and 60% thresholds selected 9, 18 and 26 features, respectively. For CIC-ToN-IoT dataset, a total of 10, 20 and 30 features were selected with the thresholds 20%, 40% and 60%, respectively. As for multivariate techniques, DISR selected a total of 14 features in both CIC-IDS2017 and CSE-CIC-IDS2018 dataset, and 16 features for CIC-ToN-IoT dataset. Furthermore, a total of 13 and 12 features were selected with CON technique in CIC-IDS2017 and CIC-ToN-IoT datasets, and only 6 features in CSE-CIC-IDS2018 dataset. Additionally, the features chosen by univariate and multivariate FS filters are listed in Table 16 and 17 in Appendix A.

Considering the optimal classifiers written in bold in Tables 10, 11 and 12, Fig. 9 illustrates the feature subsets

used for training these three optimal classifiers. As it can be seen, there are four common features among these feature subsets which are: “Flow Duration” or the time between the reception of the first and the last packet of the flow, “Init Win bytes backward” which stands for number of bytes transferred in the initial window in the backward direction, “Init Win bytes forward” or the number of bytes transferred in initial window in the forward direction, and “Port” either the destination or source ports. This means that Flow Duration, Init Win bytes backward, Init Win bytes forward and ports features were the most relevant features IDS classification regardless the environment and attack scenarios.

6.6 Accuracy results comparison

In this subsection, we present a comparative analysis of the experimental results with those of some existing works in the field. To validate and facilitate an overall understanding of the contributions and effectiveness of our proposed approach, Table 14 shows the performance results of the best combinations of classifiers and filters in the present study compared with those reported in previous research, focusing on two pivotal aspects: accuracy and the number of features selected. Table 14 provides a valuable reference point for assessing the improvements and progress made by our proposed approach. Through this comparative analysis, we aim to provide a clear and objective perspective on how our methodology excels in the context of IDSs. In addition, the majority of the studies presented in Table 14 used black box classifiers such as RF, SVM, and ANN.

Table 14 gives a succinct overview of the performance results obtained in our study compared with previous studies. This comparative analysis highlights a notable achievement: our proposed methodology consistently outperforms existing approaches in terms of accuracy, reaffirming its effectiveness as a powerful tool for IDSs. The results presented in Table

Table 14 Accuracy results comparison with previous studies

Authors and year of publication	Dataset	Technique	# Of features	Best accuracy (%)
Our study	CIC-IDS2017	XGBCON and RFXON	13	99.92
	CSE-CIC-IDS2018	RFCON and XGBRF	06	99.99
	CIC-ToN-IoT	XGBDISR and RFDISR	16	99.22
Uzun et al. [58]	NSL-KDD	RF + RL	20	99.21
Nutan et al. [59]	NSL-KDD	Ensemble approach based on majority vote	12	98.46
	AWID		8	99.52
Zhou et al. [60]	NSLKDD	CFS-BA	10	99.81
	CIC-IDS2017		13	99.89
Kurniabudi et al. [26]	CIC-IDS2017	RF + Information gain	22	99.86
Aksu et al. [27]	CIC-IDS2017	RF + Fisher score	30	99.97
Ghosh et al. [31]	CSE-CIC-IDS2018	hybrid FS approach (filter and wrapper approach)	11	99.82
Linhares et al. [61]	SimpleWeb site	SDNTruth based on the entropy and Z-test	04	97.41

14 provide convincing evidence of the progress made by our approach, paving the way for more efficient and streamlined IDSs, while mitigating the complexity and resource requirements associated with a reduced feature set.

7 Threats to validity

To ensure the reliability of this study, it is important to establish the scope of validity of the findings reached in this study. In this work, we were not able to process every dataset for binary and multiclass classification task, because of their huge size, and the training time that each model takes in the parameters tuning step. Moreover, the cybersecurity domain is a very critical one, using only one common feature set is neither enough to test the generalizability of features nor to define its trustworthiness.

The FS techniques were applied to four black-box classifiers. Constructing and comparing with other types of models such as white-box models will help to select the best combination to minimize the number of features and generalize their efficiency on network datasets; therefore, this can be used and discussed in future work.

If a model is not highly interpretable, the field may not be legally allowed to use its knowledge to make process changes. Future work will focus on applying interpretability techniques to these models to measure the importance of the selected feature subset and improve their reliability, which can help domain experts understand or discover which features can be useful for intrusion detection.

8 Conclusion and future works

This paper studied the effect of FS by comparing five univariate filters and three multivariate filters. To this end, the relevant features of three datasets related to IDS were selected using CON and DISR multivariate filters, and RL, K, and C univariate filters with 40% and 60% thresholds. SVM, MLP, RF and XGB classifiers were trained with the entire and reduced feature sets and evaluated using the 10-fold cross-validation method. The findings of this study suggested that:

- (RQ1): Even if FS techniques have no impact on model performance, they can nevertheless be useful to reduce the computational complexity of models, improve their interpretability or speed up their learning time.
- (RQ2): The occurrence of univariate FS techniques slightly exceeds that of multivariate techniques. One possible explanation for this is that univariate filters generally contain more features than multivariate filters, giving them an advantage in appearing more frequently in the initial clusters as they contain more information, which

means that univariate filters seem to outperform multivariate filters across all datasets.

- (RQ3): The use of XGB and RF models with CON and DSIR techniques can give a very robust performances in terms of accuracy and computation time, even with a small number of features, which can be useful for the interpretation of the models used.
- (RQ4): In this experiment, we demonstrated that four features namely "Flow Duration", "Init Win bytes backward", "Init Win bytes forward", and "Port" are involved in the classification of intrusions in the three environments used, which can be useful for researchers or developers of IDSs.

As future work, we suggest investigating new approaches that will rely on simulation by creating a new topology and generating a new dataset. This can serve as a valuable reference point for future research in the field of cybersecurity. Furthermore, the ongoing work aims to investigate interpretability techniques to be highly useful in understanding the black box model's behavior and increasing reliability which can help to understand or discover hidden patterns and interpret the classification decisions for IDSs. Then, allow a better understanding of how each black-box model behaves, which features it relies on most, and how each feature affects the final decision.

Author Contributions HZ and AI designed the experiments to empirically evaluate the impacts of feature selection in IDS. HZ implemented all the experiments. HZ and AI analysed and interpreted the results. All the authors prepared, reviewed and edited the paper.

Data Availability This article uses public datasets, which are publicly available online at: (<https://www.unb.ca/cic/datasets/index.html>) and (https://staff.itee.uq.edu.au/marius/NIDS_datasets/).

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Appendix

See Tables 15, 16 and 17.

Table 15 Description list of features designed by CICFlowMeter tool

ID	Feature	Description
1	Source IP	The source IP address
2	Destination IP	The destination IP address
3	Dst Port	The number of port source
4	Srs Port	The number of port destination
5	Protocol	Type of protocol, e.g TCP, UDP
6	Timestamp	the time of sending the flow
7	Flow Duration	Record total duration
8	Tot Fwd Pkts	Total packets in the forward direction
9	Tot Bwd Pkts	Total packets in the backward direction
10	TotLen Fwd Pkts	Total size of packet in forward direction
11	TotLen Bwd Pkts	Total size of packet in backward direction
12	Fwd Pkt Len Max	Maximum size of packet in forward direction
13	Fwd Pkt Len Min	Minimum size of packet in forward direction
14	Fwd Pkt Len Mean	Average size of packet in forward direction
15	Fwd Pkt Len Std	Standard deviation size of packet in forward direction
16	Bwd Pkt Len Max	Maximum size of packet in backward direction
17	Bwd Pkt Len Min	Minimum size of packet in backward direction
18	Bwd Pkt Len Mean	Mean size of packet in backward direction
19	Bwd Pkt Len Std	Standard deviation size of packet in backward direction
20	Flow Bytes/s flow	byte rate that is number of packets transferred per second
21	Flow Pkts/s flow	packets rate that is number of packets transferred per second
22	Flow IAT Mean	Average time between two flows
23	Flow IAT Std	Standard deviation time two flows
24	Flow IAT Max	Maximum time between two flows
25	Flow IAT Min	Minimum time between two flows
26	Fwd IAT Tot	Total time between two packets sent in the forward direction
27	Fwd IAT Mean	Mean time between two packets sent in the forward direction
28	Fwd IAT Std	Standard deviation time between two packets sent in the forward direction
29	Fwd IAT Max	Maximum time between two packets sent in the forward direction
30	Fwd IAT	Min Minimum time between two packets sent in the forward direction
31	Bwd IAT	Tot Total time between two packets sent in the backward direction
32	Bwd IAT Mean	Mean time between two packets sent in the backward direction
33	Bwd IAT	Std Standard deviation time between two packets sent in the backward direction
34	Bwd IAT	Max Maximum time between two packets sent in the backward direction
35	Bwd IAT	Min Minimum time between two packets sent in the backward direction
36	Fwd PSH	Flags Number of times the PSH flag was set in packets travelling in the forward direction (0 for UDP)
37	Bwd PSH	Flags Number of times the PSH flag was set in packets travelling in the backward direction (0 for UDP)
38	Fwd URG	Flags Number of times the URG flag was set in packets travelling in the forward direction (0 for UDP)
39	Bwd URG	Flags Number of times the URG flag was set in packets travelling in the backward direction (0 for UDP)
40	Fwd	Header Len Total bytes used for headers in the forward direction
41	Bwd	Header Len Total bytes used for headers in the forward direction
42	Fwd Pkts/s	Number of forward packets per second
43	Bwd Pkts/s	Number of backward packets per second
44	Pkt Len Min	Minimum length of a flow

Table 15 continued

ID	Feature	Description
45	Pkt Len Max	Maximum length of a flow
46	Pkt Len Mean	Mean length of a flow
47	Pkt Len Std	Standard deviation length of a flow
48	Pkt Len Var	Minimum inter-arrival time of packet
49	FIN Flag Cnt	# Of packets with FIN
50	SYN Flag Cnt	# Of packets with SYN
51	RST Flag Cnt	# Of packets with RST
52	PSH Flag Cnt	# Of packets with PUSH
53	ACK Flag Cnt	# Of packets with ACK
54	URG Flag Cnt	# Of packets with URG
55	CWE Flag Count	# Of packets with CWE
56	ECE Flag Cnt	# Of packets with ECE
57	Down/Up Ratio	Download and upload ratio
58	Pkt Size Avg	Average size of packet
59	Fwd Seg Size Avg	Average size observed in the forward direction
60	Bwd Seg Size Avg	Average size observed in the backward direction
61	Fwd Byts/b Avg	Average number of bytes bulk rate in the forward direction
62	Fwd Pkts/b Avg	Average number of packets bulk rate in the forward direction
63	Fwd Blk Rate Avg	Average number of bulk rate in the forward direction
64	Bwd Byts/b Avg	Average number of bytes bulk rate in the backward direction
65	Bwd Pkts/b Avg	Average number of packets bulk rate in the backward direction
66	Bwd Blk Rate Avg	Average number of bulk rate in the backward direction
67	Subflow Fwd Pkts	The average number of packets in a sub flow in the forward direction
68	Subflow Fwd Byts	The average number of bytes in a sub flow in the forward direction
69	Subflow Bwd Pkts	The average number of packets in a sub flow in the backward direction
70	Subflow Bwd Byts	The average number of bytes in a sub flow in the backward direction
71	Init Fwd Win Byts	Number of bytes sent in initial window in the forward direction
72	Init Bwd Win Byts	# Of bytes sent in initial window in the backward direction
73	Fwd Act Data Pkts	# Of packets with at least 1 byte of TCP data payload in the forward direction
74	Fwd Seg Size Min	Minimum segment size observed in the forward direction
75	Active Mean	Mean time a flow was active before becoming idle
76	Active Std	Standard deviation time a flow was active before becoming idle
77	Active Max	Maximum time a flow was active before becoming idle
78	Active Min	Minimum time a flow was active before becoming idle
79	Idle Mean	Mean time a flow was idle before becoming active
80	Idle Std	Standard deviation time a flow was idle before becoming active
81	Idle Max	Maximum time a flow was idle before becoming active
82	Idle Min	Minimum time a flow was idle before becoming active
83	Label	Intrusion or not

Table 16 Selected features in each dataset using univariate filters (each number refers to a feature ID)

Univariate filters Datasets	Threa-shold	#.Of features	Rlieff	Pearson	Chi-2	ANOVA	Mutual information
CIC-IDS2017	20%	8	7,16,17,23, 38,49,72,74	3,7,16,17, 23,44,49,52	3,7,16,23, 48,49,52,54	3,7,16,17, 22,23,44,48	10,12,14,16, 27,48,71,72
	40%	17	7,16,17,23, 31,33,34,38, 48,49,51,52,54 ,57,71,72,74	3,7,13,16, 17,22,23,27, 33,34,43,44,48, 54,49,52,71	3,7,16,17, 22,23,27,33, 34,36,48,49,52, 53,54,72,80	3,7,13,16, 17,22,23,33, 34,44,49,48,52, 53,54,72	3,7,10,12, 14,16,20,21, 22,27,31,34,40, 43,48,71,72
	60%	25	3,7,16,17,21, 23,25,27,30,31, 32,33,34,38,48, 49,51,52,53,54, 57,71,72,74,80	3,7,12,13,14, 16,17,20,21,22, 23,27,33,34,35, 36,43,44,48,54, 49,52,71,72,77	3,7,12,13,14, 16,17,22,23,25, 27,30,32,33,34, 35,36,44,48,49, 52,53,54,72,80	3,7,12,13,14, 16,17,21,22,23, 27,33,34,35,36, 44,49,48,52,53, 54,57,72,78,80	3,7,8,10,12,13, 14,16,20,21,22, 23,27,31,32, 34,40,43,44,48, 71,72,75,77,78
CSE-CIC-IDS2018	20%	9	16,19,38,51 3,7,12,14,15, 16,19,38,51	18,51,52,72 3,5,7,16,17, 18,51,52,72	51,52,53,72 3,5,7,16,31, 51,52,53,72	51,52,53,74 3,12,14,15,16, 51,52,53,74	18,19,46,48 3,12,14,15,16, 18,19,46,48
	40%	18	3,7,12,14,15, 16, 18,19,22, 29,38,46,48,51, 52,53,71,72	3,5,7,13,14, 15,16,17,18,19, 23,29,31,44,48, 51,52,72	3,5,7,12,15, 16,19,22,29,31, 34,36,51,54,52, 53,71,72	3,5,12,13,14, 15,16,17,18,19, 22,44,51,52, 53,71,72,74	3,7,12,14,15, 16,18,19,21,22, 29,31,34,43, 46,48,71,72
	60%	26	3,5,7,8,12, 14,15,16,17,18, 19,22,29,31, 34,36,38,46, 48,51,52,53, 54,71,72,74	3,5,7,13,14, 15,16,17,18,19, 23,28,29,31, 32,33,34,44, 46,48,51,52, 53,57,72,74	3,5,7,12,14, 15,17,16,18,19, 22,23,28,29, 31,33,34,36, 44,48,51,54, 52,53,71,72	3,5,7,12,13, 14,15,16,17,18, 19,22,24,28, 31,44,46,48, 51,52,53,57, 71,72,74	3,7,8,9,12, 14,15,16,18,19, 20,21,22,23,28, 29,31,32,33, 34,35,43, 46,48,71,72
CIC-ToN-IoT	20%	10	7,8,9,10,11,12, 13,14,16,17	3,4,5,12,14,16, 18,45,46,71	3,4,5,12,14,16, 18,45,46,71	3,4,5,12,14,16, 18,45,46,71	3,10,11,12,14, 16,18,40,41,72
	40%	20	7,8,9,10,11, 13,14,16,17	3,4,5,12,14, 18,45,46,71	3,4,5,12,13, 18,45,46,71	3,4,5,12,14, 18,45,46,71	3,4,9,10,11, 16,18,40,41,72

Table 16 continued

Univariate filters Datasets	Threa-shold	# Of features	Rlieff	Pearson	Chi-2	ANOVA	Mutual information
	60%	30	12,13,14,16,17,	16,17,18,21,22,	14,16,17,18,21,	16,17,18,21,22,	12,14,16,18,31,
			18,20,21,22,23,	24,29,43,44,45,	43,44,45,46,71,	24,29,43,44,45,	32,34,40,41,45,
			24,27,28,29,31	46,71,72,79,81,	72,74,75,78,79	46,71,72,79,81	46,71,72,79,81
			7,8,9,10,11,	3,4,5,7,10,	3,4,5,10,12,	3,4,5,7,10,	3,4,5,7,9,
			12,13,14,16,	12,13,14,16,17,	13,14,16,17,	12,13,14,16,	10,11,12,14,
			17,18,20,21,22,	18,21,22,23,	18,21,23,28,	17,18,21,22,	16,18,20,21,
			23,24,27,28,	24,27,29,43,44	33,40,43,44,	23,24,27,29,43,	23,29,31,32,27,
			29,31,32,33,	45,46,71,72,	45,46,50,66,	44,45,46,71,	34,40,41,43,45,
			34,40,41,43,	74,75,76,77,	71,72,74,75,	72,74,75,76,	46,71,72,73,
			44,45,46,50	78,79,81	76,77,78,79	77,78,79,81	74,79,81

Table 17 Selected features in each dataset using multivariate filters (each number refers to a feature ID)

Multivariate filters Datasets	CFS	# Of features	CSS	# Of features	DISR	# Of features
CIC-IDS2017	3,48,51,71,72	5	3,7,20,21,22, 23,25,30,33,	40,48,71,72	3,8,10,12,13, 14,23,45,49,71,	72,75,77,78
CSE-CIC-IDS2018	3,18,54,77	4	3,7,8,24,71,72		3,5,36,52,51, 53,54,71,72,44,	45,48,57,74
CIC-ToN-IoT	3,73,81	3	18,20,21,22, 24,29,40,43,	51,71,79,81	3,4,5,7,9,10, 11,12,13,64,65,71,	72,74,79,81

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